

Advanced Applied Mathematics III (Math 2042)

Assignment I and II (20%),

Write all necessary steps clearly and neatly

1. Solve each of the following differential equations:

a) $2ydx + (6 + 6x - y)dy = 0, \quad y(1) = 2.$

b) $x^2y' - \frac{3x^3}{x^2+1}y = 10x^3y^4$

c) $y'' + 3y' + 2y = \frac{1}{1+e^{2x}}$

2. Using **Laplace Transform**, solve the initial value problem (IVP)

$$\begin{aligned} x' - y &= e^t \\ y' + x &= \sin t \end{aligned} \quad x(0) = 1, y(0) = 0$$

3. An object moves counterclockwise along the curve $C: f(x) = 2x^2 + 2x$ beginning at the point $(1,4)$.

a) Find the position function $\mathbf{r}(t)$ that describes the motion of moving object at any time t .

b) Find the velocity and acceleration function of the moving object

c) Find the distance covered by the object when $t = 5$.